Run report: eight more draws, invariant against raw guesses at $\ell = 8$

- **Study:** `research/dreg_invariant_guess_more_20261029/`, pre-registered before any new draw was generated. No amendments.
- **Run:** 2026-10-29, 14:32 to 21:45 UTC.
- **Class:** stage diagnostic. No `S`, no rho ratio, no speed claim.

What ran

- **Draws:** `dreg_ladder`, seed 20261022, cell (22, 8), continuing the committed sequence from index 7. Its replay of draws 0–6 equals the committed draws.
- **Registered set:** the first 8 unsatisfiable draws: 8, 11, 14, 15, 17, 24, 25 and 27.
- **Arms:** the same `dreg_linear_guess` build as 2851, at `--to 4 --max-cols 140000 --check`.
  - S3: $x_{15} := x_7 + c, x_{23}, x_{24}$.
  - X3: x_7, x_{23}, x_{24}.
  - 8 branches each, stopping at the first branch that saturates degree 4 without refuting.
- **Positive control:** `{7, 15, 23, 24} = 0` on every draw.
- **Limits:** `prlimit --as=15e9`, `timeout 7200`, one process at a time, on the 4-core, 16 GB container, not isolated.

Completion

- 81 processes exited 0, and none exited non-zero.
- Two runs were killed in flight (the session's background limit, then a container restart) and repeated. Their empty first attempts are in `runs/attempts/`.
- Peak RSS was 7.0–8.6 GB.

Measured values

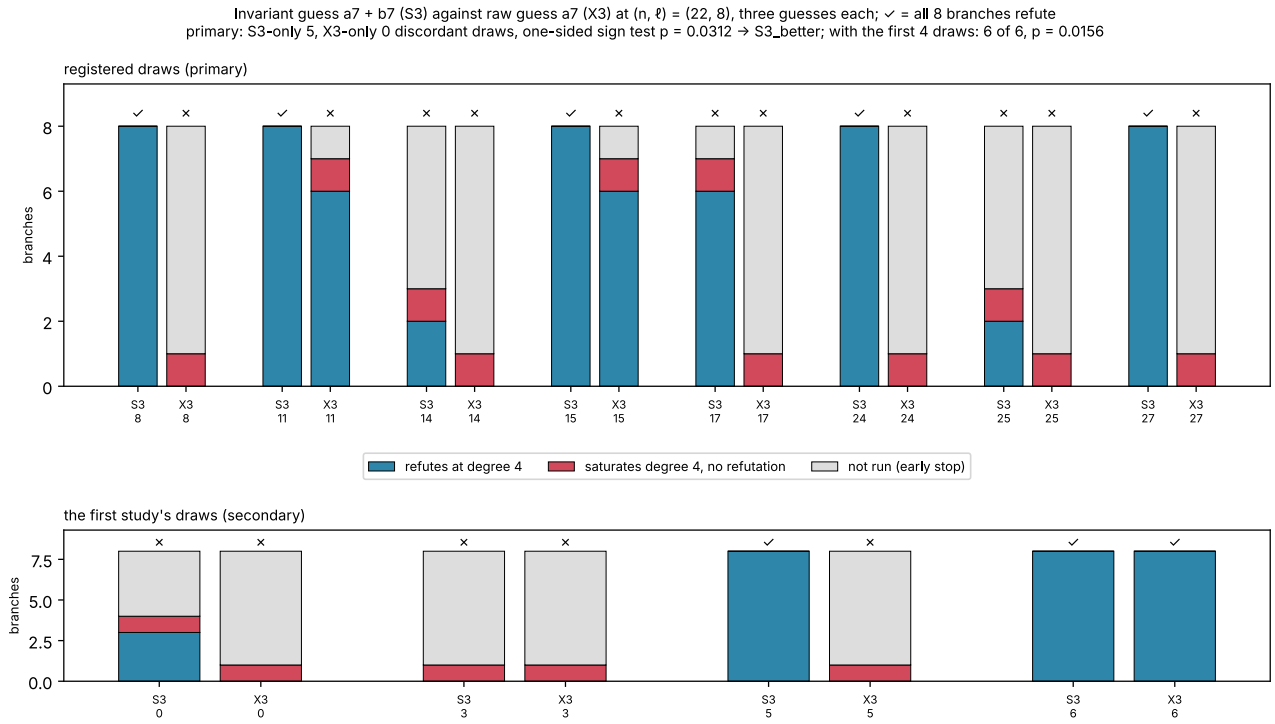


Figure 1: Branches per arm and draw: refuting at degree 4, saturating degree 4 without refutation, or not run. Top: the registered draws. Bottom: the four draws of 2851 Source: `plot.py` (layout only) on `plot-data.json` from `dreg_linear_guess_more_score --plot-data`.

- **Controls:** the replay check passes, every run passes the identity and exactness checks, and the positive control refutes on 8 of 8. Valid.
- **S3** suffices on 5 of 8 draws, and **X3** on 0 of 8.
- **Registered verdict:** S3_better. All 5 discordant draws favour S3; the one-sided exact sign test gives p = 0.031.
- **Secondary:** with 2851′s draws, 6 of 6 discordant draws favour S3, p = 0.016.